

RAY OPTICS & OPTICAL INSTRUMENTS



Ray optics is based on the behaviour of light when it travels in the form of **rays**.

Key Ideas

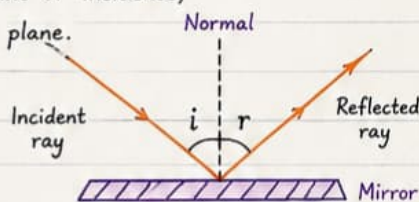
- Light travels in straight line
- Laws of reflection & refraction
- Image formation by mirrors
- Image formation by lenses
- Optical instruments

1. NATURE OF LIGHT

- Light is a form of **energy** which produces the sensation of vision.
- It travels in a **straight line** in a **homogeneous medium**.
- It **does not require any material medium** for its propagation.

2. LAWS OF REFLECTION

- The angle of incidence (i) is equal to the angle of reflection (r).
- The incident ray, reflected ray and the normal at the point of incidence, all lie in the same plane.

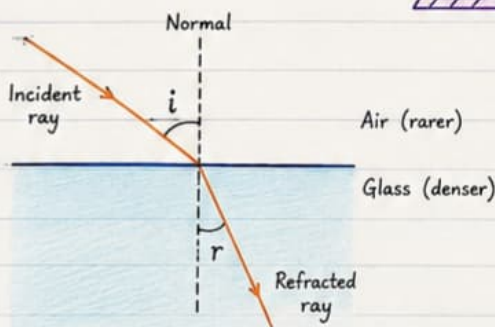


3. LAWS OF REFRACTION (Snell's Law)

- The incident ray, refracted ray and the normal at the point of incidence, all lie in the same plane.
- For a given pair of media, the ratio of sine of angle of incidence (i) to the sine of angle of refraction (r) is constant.

$$\frac{\sin i}{\sin r} = \text{constant} = n_{21}$$

where, n_{21} = Refractive index of medium 2 with respect to medium 1.

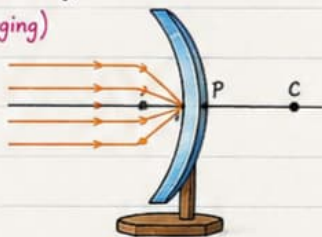


4. SPHERICAL MIRRORS

Spherical mirrors are of two types :

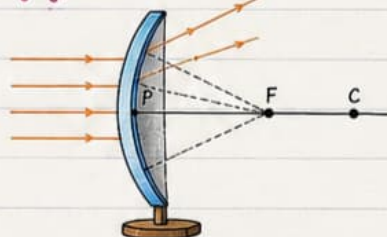
(i) Concave Mirror (Converging)

- Reflecting surface is curved inwards.
- Parallel rays converge at focus (F).
- Real and inverted images can be formed.



(ii) Convex Mirror (Diverging)

- Reflecting surface is curved outwards.
- Parallel rays diverge after reflection.
- Virtual, erect and diminished images are formed.



5. MIRROR FORMULA

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

where,

u = object distance (with sign)

v = image distance (with sign)

f = focal length (with sign)

Sign Convention (for mirrors)

- All distances measured from the pole (P).
- Distances measured in the direction of incident light are negative.
- Distances measured in the opposite direction are positive.
- Heights above principal axis are positive, below are negative.

Magnification (m)

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

h_i = height of image

h_o = height of object

$m > 1$ → enlarged image

$m = 1$ → same size image

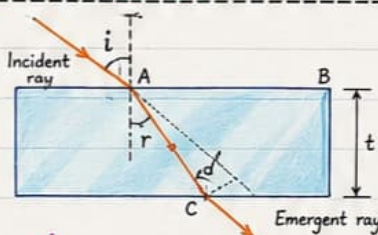
$0 < m < 1$ → diminished image

$m < 0$ → image is inverted

$m > 0$ → image is erect

6. REFRACTION THROUGH A PLANE GLASS SLAB

- A ray of light bends towards the normal on entering the slab and away from the normal on leaving the slab.
- The emergent ray is parallel to the incident ray.
- Lateral displacement (d) occurs.



Lateral Displacement (d)

The perpendicular distance between incident ray produced and emergent ray.

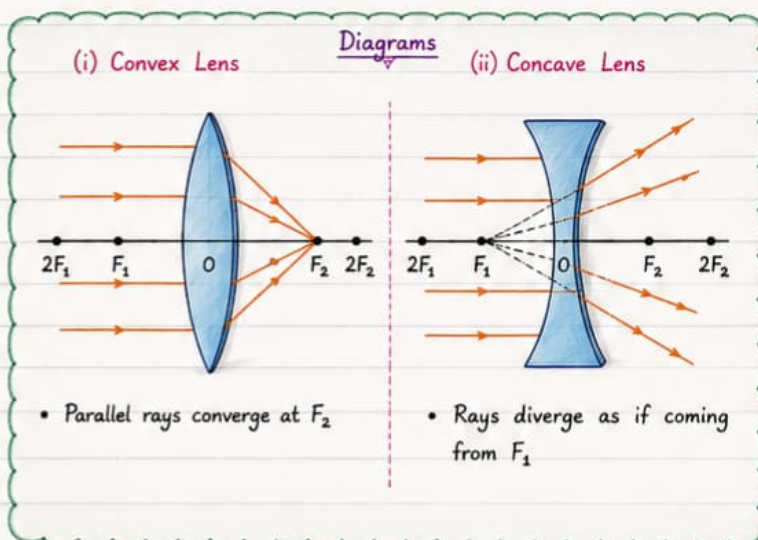
(d) increases with t and angle of incidence i

7. THIN LENS

- A lens is a transparent medium bounded by two spherical surfaces or one spherical and one plane surface.

Types of Thin Lens :

- (i) **Convex Lens (Converging Lens)**
 - Thicker at the centre, thinner at the edges.
 - Converges a parallel beam of light at principal focus (F_2).
- (ii) **Concave Lens (Diverging Lens)**
 - Thinner at the centre, thicker at the edges.
 - Diverges a parallel beam of light. The rays appear to come from principal focus (F_1).



8. LENS FORMULA (For Thin Lens)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

where,

u = object distance (with sign)

v = image distance (with sign)

f = focal length (with sign)

Sign Convention

- All distances measured from optical centre (O).
- Towards right $\rightarrow +ve$
- Towards left $\rightarrow -ve$
- Height above principal axis $\rightarrow +ve$
- Height below principal axis $\rightarrow -ve$

9. MAGNIFICATION

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

where,

m = magnification

h_i = height of image

h_o = height of object

Nature of Image

- $m > 0 \rightarrow$ image is virtual and erect
- $m < 0 \rightarrow$ image is real and inverted
- $|m| > 1 \rightarrow$ image is enlarged
- $|m| < 1 \rightarrow$ image is diminished
- $|m| = 1 \rightarrow$ image is same size

10. IMAGE FORMATION BY A CONVEX LENS

S.No.	Position of Object	Ray Diagram	Position of Image	Nature of Image	Size of Image
1.	At infinity ($u = -\infty$)		At F_2 ($v = +f$)	Real & Inverted	Highly diminished (point sized)
2.	Beyond $2F_1$ ($u < -2f$)		Between F_2 and $2F_2$	Real & Inverted	Diminished
3.	At $2F_1$ ($u = -2f$)		At $2F_2$	Real & Inverted	Same size
4.	Between F_1 and $2F_1$ ($-2f < u < -f$)		Beyond $2F_2$	Real & Inverted	Enlarged
5.	At F_1 ($u = -f$)		At infinity ($v = +\infty$)	Real & Inverted	Highly enlarged
6.	Between O and F_1 ($u > -f$)		On the same side of lens ($v < 0$)	Virtual & Erect	Enlarged

Key Points

- For a convex lens, real images are always inverted.
- A convex lens can form both real and virtual images.
- A concave lens always forms a virtual, erect and diminished image for any position of the object.

Remember!

Light follows the laws of reflection and refraction while travelling through different media. Ray diagrams help us predict the path of light and the nature, position and size of images.

11. IMAGE FORMATION BY A CONCAVE LENS

S.No.	Position of Object	Ray Diagram	Position of Image	Nature of Image	Size of Image
1.	At infinity ($u = -\infty$)		At F_1 ($v = -f$)	Virtual & Erect	Highly diminished (point sized)
2.	Between infinity and $2F_1$ ($u < -2f$)		Between O and F_1 ($-f < v < 0$)	Virtual & Erect	Diminished
3.	At $2F_1$ ($u = -2f$)		Between O and F_1 ($-f < v < 0$)	Virtual & Erect	Diminished
4.	Between F_1 and O ($-f < u < 0$)		Between O and F_1 ($-f < v < 0$)	Virtual & Erect	Diminished
5.	At O ($u = 0$)		At O ($v = 0$)	Virtual & Erect	Same size

Key Points

- A concave lens always forms virtual, erect and diminished images regardless of the position of the object.
- Image is always formed on the same side of the lens as the object.
- Concave lens is used to correct short sightedness (myopia).



Power of Lens (P)

$$P = \frac{1}{f} \text{ (in metres)}$$

Unit : Diopetre (D)

$P > 0 \rightarrow$ Convex lens

$P < 0 \rightarrow$ Concave lens

12. COMBINATION OF THIN LENSES (IN CONTACT)

- When two or more thin lenses are placed in contact, their equivalent focal length (f_{eq}) is given by :

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \dots$$

- If the lenses are separated by small distances (d), the formula becomes :

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \text{ (for two lenses)}$$

(d is the distance between the two lenses)

Examples :

- A convex lens of $f_1 = 20$ cm is in contact with another convex lens of $f_2 = 30$ cm. Find f_{eq} .

$$\frac{1}{f_{eq}} = \frac{1}{20} + \frac{1}{30} = \frac{3+2}{60} = \frac{5}{60}$$

$$f_{eq} = 12 \text{ cm}$$

$f_1 = 20 \text{ cm}$ $f_2 = 30 \text{ cm}$

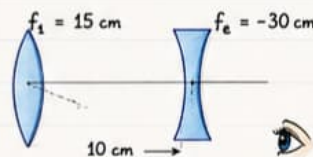


- A convex lens of $f_1 = 15$ cm is placed 10 cm apart from a concave lens of $f_2 = -30$ cm. Find f_{eq} .

$$\frac{1}{f_{eq}} = \frac{1}{15} + \frac{1}{-30} - \frac{10}{15 \times -30}$$

$$= \frac{2-1+2}{30} = \frac{3}{30} = \frac{1}{10}$$

$$f_{eq} = 10 \text{ cm}$$

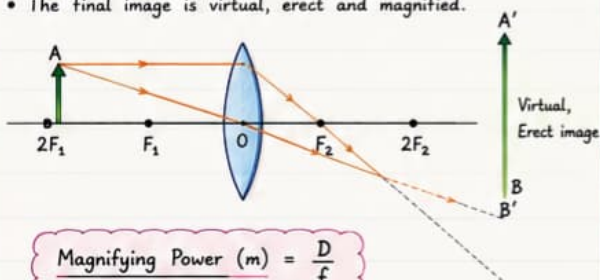


13. OPTICAL INSTRUMENTS

(ii) SIMPLE MICROSCOPE



- It consists of a single convex lens of short focal length.
- The object is placed between O and F .
- The final image is virtual, erect and magnified.



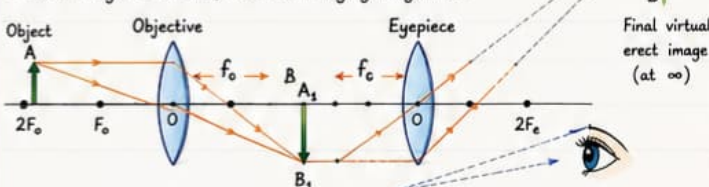
$$\text{Magnifying Power (m)} = \frac{D}{f}$$

Where, D = least distance of distinct vision (≈ 25 cm)
 f = focal length of the lens

(ii) COMPOUND MICROSCOPE



- It consists of two convex lenses:
 - (1) **Objective** (f_o) $\rightarrow$ small focal length, forms real, inverted and magnified image.
 - (2) **Eyepiece** (f_e) $\rightarrow$ acts as a magnifying glass.
- Final image is virtual, erect and highly magnified.



$$\text{Total Magnifying Power (m)} = \frac{L}{f_o} \times \frac{D}{f_e}$$

Where, L = length of the microscope tube

Quick Revision



- Light travels in straight line.
- Convex lens converges, concave lens diverges.
- Power of lens $P = \frac{1}{f}$ (metres)
- Microscopes & telescopes help us see very small or far objects clearly.
- Real image can be obtained on screen, virtual image cannot.



14. REFLECTION AT SPHERICAL SURFACES

- The mirror formula is valid for both concave and convex mirrors.
- Distances are measured from the pole (P).
- Cartesian sign convention is used.

Mirror Formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

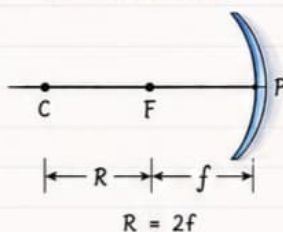
where,

f = focal length

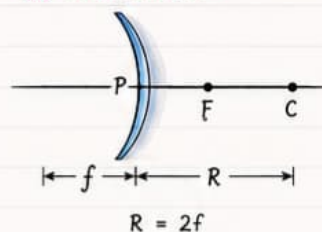
v = image distance

u = object distance

(i) Concave Mirror



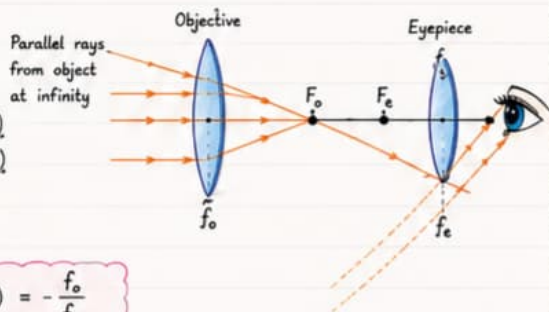
(ii) Convex Mirror



15. OPTICAL INSTRUMENTS - TELESCOPES

(i) Astronomical Telescope

- Used to observe distant objects (stars, planets).
- Objective: large focal length (f_o).
- Eyepiece: small focal length (f_e).
- Final image: virtual, inverted and highly magnified.

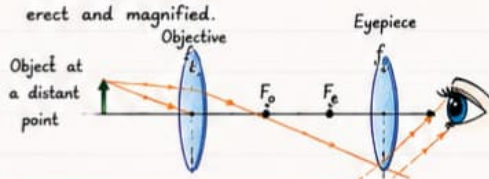


$$\text{Magnifying Power (m)} = -\frac{f_o}{f_e}$$

(Image is inverted)

(ii) Terrestrial Telescope

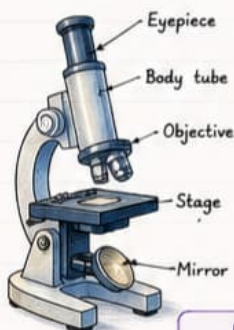
- Used to observe distant objects on Earth.
- Objective and eyepiece have shorter focal lengths.
- Final image: virtual, erect and magnified.



$$\text{Magnifying Power (m)} = \frac{f_o}{f_e}$$

(Image is erect)

16. MICROSCOPE (COMPOUND MICROSCOPE)



- Consists of two convex lenses: objective (f_o) and eyepiece (f_e).
- Object is placed just beyond focus of objective.
- Intermediate image is formed near the first focal plane of eyepiece.
- Final image is virtual, highly enlarged and inverted.

Magnifying Power (m)

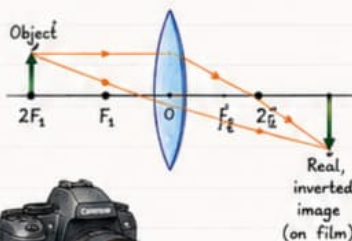
$$= -\frac{L}{f_o} \times \frac{D}{f_e}$$

where, L = length of tube
 D = least distance of distinct vision (~ 25 cm)

17. CAMERA AND HUMAN EYE

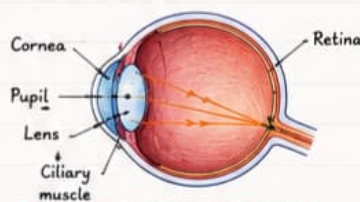
(i) Camera

- A camera uses a convex lens to form a real, inverted and diminished image on the film.
- The object must be beyond $2F$.



(ii) Human Eye

- Eye acts like a camera.
- Cornea + lens $\rightarrow$ convex lens
- Retina acts as the screen.
- Image formed is real, inverted and diminished.



Power of Accommodation

- The ability of eye lens to change its focal length with the help of ciliary muscles.



18. SOME IMPORTANT FORMULAS

- Mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
- Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$
- Magnification (m): $m = \frac{h_i}{h_o} = -\frac{v}{u}$
- Power of lens: $P = \frac{1}{f}$ (f in metre)
- Combination of lenses (in contact):
 $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \dots$
- Combination of thin lenses (separated by d):
 $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$ (for two lenses)

19. COMMON TERMS (AT A GLANCE)

Term	Meaning	Concave Mirror	Convex Mirror	Convex Lens	Concave Lens
Real image	Formed by actual rays	Possible	Not possible	Possible	Not possible
Virtual image	Formed by apparent rays	Possible	Possible	Possible	Possible
Erect image	Same orientation	Possible	Always	Possible	Possible
Inverted image	Opposite orientation	Possible	Not possible	Possible	Not possible
Magnification	Size of image size of object	May be >1 , $=1$, <1	Always <1	May be >1 , <1	Always <1

Quick Revision

- Light travels in straight lines.
- Reflection & refraction govern the path of light.
- Lenses can converge (convex) or diverge (concave).



- Mirrors form images by reflection.
- Lenses form images by refraction.
- Optical instruments help us see far, small or invisible objects.

Understand concepts
Clearly $\rightarrow$ Draw ray diagrams
Neatly $\rightarrow$ Solve numericals
Smartly! ✨

